Yuan-Jyun Lee

Taipei, Taiwan
xu3m06rmp4@gmail.com — +886 975-706-016 — Personal Page

RESEARCH INTERESTS

Operator Algebras, Non-commutative Dynamical Systems, Quantum Markov semi-groups, Ergodic Properties

EDUCATION

National Yang Ming Chiao Tung University (NYCU) — Taiwan
B.S. in Department of Applied Mathematics — Fall 2022 — Present
Cumulative GPA: 4.01/4.30

National Yang Ming Chiao Tung University (NYCU) — Taiwan
B.S. in Department of Electronics and Electrical Engineering — Fall 2023 — Present

Honors

Academic Excellence Awards — Hsinchu, Taiwan
Awarded for outstanding academic performance — Spring 2022, Fall 2023

College of Science Dean Award — Hsinchu, Taiwan
For ranking in the top 10% of the department in my first three years of undergraduate study. — 2023

Zhu Shun-Yi Cooperative Scholarship — Hsinchu, Taiwan
Scholarship for academic excellence and contribution to research — 2025

Mathematics Presidential Award — Hsinchu, Taiwan
Awarded by NYCU Applied Mathematics Department for outstanding performance — Spring 2022, Fall 2022, Fall 2023

Excellence in Undergraduate Research — Hsinchu, Taiwan
Awarded for outstanding results in undergraduate research projects — 2025

PROJECTS

Research reading group on QMS theory
Hsinchu, Taiwan
Independent research in QMS theory — Spring 2025 — Present
Advisor: Cheng-Fang Su

- Studying mathematical foundations of quantum Markov semigroups and open quantum system dynamics.
- Reviewing topics including completely positive trace-preserving (CPTP) maps, GKSL theorem, Fregiro's theorem and functional calculus for generators.
- Investigating connections between Markovian and non-Markovian quantum dynamics, including process tensor formulations.

Research reading group on ergodic theory
Hsinchu, Taiwan
Studied foundational results in ergodic theory — Summer 2025
Advisor: Xiang Fang

- Studied fundamental topics in ergodic theory, including measure-preserving transformations and ergodic theorems.
- Prepared detailed lecture notes and expository materials for instructional use.
- Primary reference: Walters, *An Introduction to Ergodic Theory* (2000).

Undergraduate Summer Research Program, TWSIAM
Hsinchu, Taiwan
A Study on Trace Inequalities Derived from Relative Entropy — Fall 2024 — Spring 2025
Advisor: Cheng-Fang Su, Hao-Wei Huang

- Studied mathematical foundations of quantum information theory, focusing on entropy inequalities.
- Established preliminary trace inequality results under structural constraints on density matrices, contributing to ongoing research in the area.

Undergraduate Summer Research Program, TWSIAM
Hsinchu, Taiwan
A research on finite time Kuramoto model Summer 2023
Advisor: Ming-Cheng Shiue

- Studied synchronization phenomena in the Kuramoto model.
- Improved bounds on convergence rates for finite-time variants of the model.

CONFERENCE

Oral Presentation, *Finite time convergence of Kuramoto model*
2024 NTNU Fall Applied Mathematics Student Conference
Recipient of Best Presentation Award.

SELECTED COURSES

Graduate Coursework

- Functional Analysis
- Operator Theory II
- Real Analysis I, II(Measure Theory)
- Quantum Mechanics
- Algebraic Graph Theory
- Stochastic Process
- Fluid Dynamics

TEACHING EXPERIENCE

Fall 2025: Calculus I
Fall 2025: Probability
Fall 2025: Fundamental Mathematics

OTHER EXPERIENCES

President, Mathematics Research Club Hsinchu, Taiwan
Organized biweekly research meetings, managed club activities, and facilitated academic discussions Fall 2025 – Present

Founder, Department Football (Soccer) Team Hsinchu, Taiwan
Established and organized the NYCU Mathematics Department football (soccer) team Fall 2022 – Present

Captain, Department Basketball Team Hsinchu, Taiwan
Led team training and coordinated inter-departmental competitions 2023

ENGLISH

TOEFL(ibt): 5.5 (100) (overall score)

Listening: 5.5 — Reading: 5.5
Speaking: 3.5 — Writing: 5
Test date: Mar. 2026

SKILLS

- **Programming:** C++, Python, Matlab
- **Languages** Mandarin, English